LEF MODEL · CHAPTER 13

# Chapter 13: The Poles of the Moon — Why the Poles Are Crater-Crowded, Six Hypotheses Rejected and the Triaxial Shape Recovered

Based on Paper 07 — The Poles: Why the Poles Are Crowded (Parts A–C.2: the clean count, the 6 rejected hypotheses and the geometry of flattening)

01 · THE MYSTERY

## The mystery: why are both poles of the Moon crowded with large craters — and even more so with the giant ones?

When all the large craters (20 km and more) are counted in the precise global catalogue of Robbins (2019), a sharp two-pole pattern appears that no one had explained before: **the two polar caps (radius 30° around the north pole and the south pole) are much more crowded with craters than the equatorial belt (15°S–15°N)**!

- **In the raw count:** the polar caps carry **219 craters** per million km² (227 in the north and 211 in the south) against **166** in the equatorial belt — a ratio of **1.32** (and for craters 40 km and more the ratio jumps to **1.51**).
- **The suspicion of a geological bias — and the solution in the "clean count":** the sceptic will at once argue that the equator is rich in young basalt maria (22% of its area against 1% at the poles) that erased old craters, while the south pole is cut by the South Pole–Aitken basin (SPA, which covers 24% of the polar caps). But when both the maria and the SPA basin are removed from

both sides (**the clean count**), the polar excess does not vanish — it settles at **1.20 ± 0.05** (305 in the south and 300 in the north in the 75°–90° band), and climbs consistently as the crater diameter grows: from **1.09** at 20–30 km to **1.53 ± 0.19** at 60–100 km and **1.84 ± 0.38** at 100–300 km!

02 · AN IMAGE

## An image: a round balloon squeezed between two palms and then released

Why can no rain of asteroids from outer space (nor an even rain of returning material from the Earth) create a symmetric two-pole concentration around the Moon's rotation axis? One image solves the riddle at once:

- **A dotted balloon squeezed between two palms (the flattening stage):** imagine a round balloon covered with dots (craters) at an even density. Now press it hard from above and below (the centrifugal force of the fast 5.5-hour rotation) and pull it sideways toward the Earth (Roche's tidal forces). The balloon becomes a flattened, elongated body like **"a pita that is also an egg"** (axis ratio a : b : c ≈ 1.8 : 1.4 : 1): its polar regions widen and stretch sideways into large, flat surfaces on which the craters land at an even density!
- **Releasing the hands and rounding inward (the stage of volume collapse):** the moment the Moon's centre of mass crosses the corotation line (15,818–16,399 km) and the stretching is released while it is still attached to the Earth, the Moon's self-gravity pulls it back to a spherical shape while its volume contracts (from a density of 1,700 to 2,600 kg/m³). The balloon springs back into a sphere: **the wide, flat surfaces of the poles converge inward toward the two centres of the poles** (all the lines of longitude retreat toward the poles!), and the craters sitting on the stiff crust are crowded into a much smaller area without losing their original size!

03 · SCIENCE AND MODEL

## Science and model: separating the polar pattern from the longitude pattern, and rejecting the 6 alternative

# hypotheses

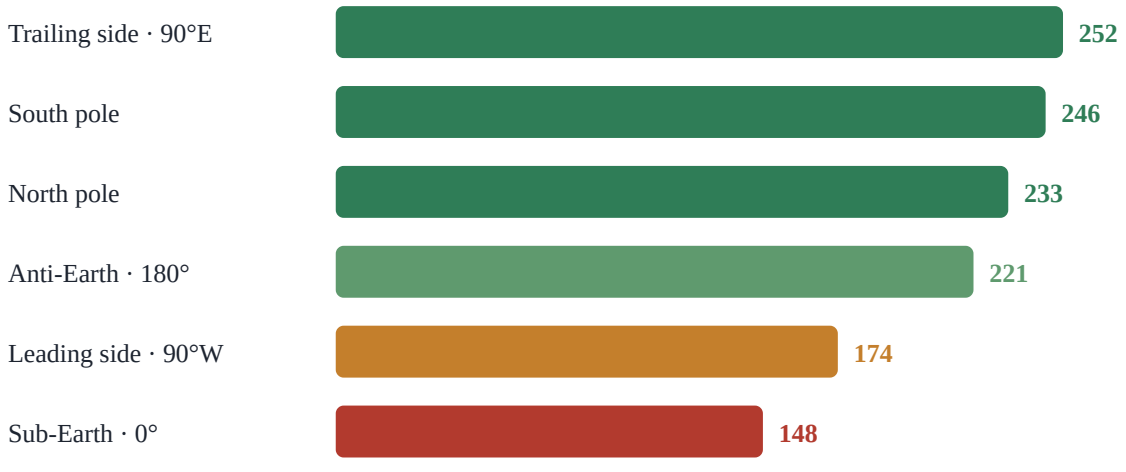
Parts A and B of Paper 07 scan the Moon's face in 12 latitude bands, 12 longitude sectors, six nested caps around each pole (10°–60°), and six caps around the six axis ends, and systematically test **six alternative hypotheses** before reaching a verdict:

1. **A full separation between the latitude pattern and the longitude pattern (§A.1–A.5):** along the lines of longitude there is a completely separate asymmetry between the trailing side (90°E, densest: 252 in the 30° cap) and the point beneath the Earth (0°, sparsest: 148). By contrast, every latitude band circles the Moon through 360° and so averages all the longitudes equally — so the polar excess is a completely independent geometric phenomenon!

2. **The SPA basin is limited to the inner core (0°–10°) (ring study §A.10):** measuring the density in rings around the centre of SPA (53°S, 191°E) reveals that the shortage of craters is not uniform across the whole basin, but concentrated as a "step" in the inner core 0°–10° only (**132 craters** per million km², a ratio of 0.59 against 222 outside the rim!), while in the 10°–39.57° rings the density jumps at once to 193–202. Removing SPA brings the two poles into full statistical equality (305 in the south against 300 in the north in the 75°–90° band; 280 against 270 in the 20° cap).

3. **Why were all six alternative hypotheses rejected (§B.1–B.6)?**
    - 4. **(B.1) Preservation in the ancient highlands:** fails the mask test and the shape test — even after removing the maria and SPA an excess of 1.20 ± 0.05 remains, the poles (233–246) are denser than the ancient far-side crown (221), and at the clean north pole there is an inward convergence (189 at 60° $\rightarrow$ 300 at 15°).
    - 5. **(B.2) The flattening itself (fast rotation):** the opposite sign! Pressing on the poles stretches their area outward and thins out the craters ($\rho_{\text{polar}} = (c/a)^2 \rho_{\text{eq}} = 0.59 \times$ ).

- 6 **(B.3) Roche tidal stretching:** the opposite sign! Stretching the axis facing the Earth shortens the polar axis but stretches the polar area outward (0.83×) and actually compresses the point beneath the Earth (1.38×).
- 7 **(B.4) Stronger gravity at the pole:** the opposite sign during rotation! On a flattened body spinning in 5.77 hours, the pole is energetically higher by $\Delta H = \frac{1}{2}\Omega^2 R^2/g \approx 120$ km than the equator (a "mountain top"), so material slides off it outward toward the equator — and only *after* the rotation is released does the pole become an energy valley!
- 8 **(B.5) Drainage / a point vortex at the pole:** fails the shape test — in the north the density levels off on a broad plateau (300 at 15° and 298 at 10°) with no topographic pit, because rotational forces create **a symmetric ring** 90° from the Earth line, not a singular point.
- 9 **(B.6) A uniform (isotropic) change of volume:** a uniform spherical contraction changes all distances in the same ratio and so leaves the pole/equator ratio exactly equal to 1.00.




The clean density of craters 20 km and more across (per million km²) in 30° caps around the six axis ends: from the minimum beneath the Earth to the poles and the trailing side.

The proof of the density at peak flattening ($\rho \approx 1{,}700$ kg/m³) from two independent tests (§C.1)

(1) **The Maclaurin spheroid:** for a rotation period of 5.5 hours, the measured equatorial-to-polar axis ratio (b/c ≈ 1.41) is obtained exactly at a density of **1,700 kg/m³** (below 1,070 kg/m³ there is no equilibrium and the body breaks apart).
(2) **The area-closure limit:** had the peak density been higher than **1,800 kg/m³**, the volume contraction during the rounding would have been too small, and the ends of the Earth axis would have been forced to expand instead of closing!

04 · THE MECHANISM IN WORDS

## The mechanism in words: the geometry of "the pita that is also an egg" and recovering the shape without tuning parameters

Parts B.7 and C.1–C.2 of Paper 07 show how the combination of the fast rotation with the Earth's tidal stretching created a well-defined triaxial body (a : b : c ≈ 1.8 : 1.4 : 1), and how measuring the craters around the six axis ends recovers its original shape precisely:

1. **The three axes of the attached body (§A.9, §C.1–C.2):** while the young Moon was attached to the tip of the pear and locked in a synchronous 5.5-hour rotation around the Earth, two distortions acted on it at once:
    - 2 **The "pita" part (b/c ≈ 1.41):** the fast rotation about the polar axis (c) flattened the poles and widened the equator (a Maclaurin spheroid at a density of ~ 1,700 kg/m³).
    - 3 **The "egg" part (a/c ≈ 1.8):** the Earth's gravitational pull inward on the near side against the centrifugal force outward on the far side stretched the body along the Earth axis (a, from 5,585 × 4,628 × 3,191 km at peak flattening to a spherical diameter of 3,778 km at detachment!).

4. **The ranking across the six axis ends (the six-axis study, §A.9):** measuring the crater density (20+ km) in six equal 30° caps (after removing the maria and the 20 large basins) reveals a clear triaxial ranking:
    - 5. **(1) The point beneath the Earth (0°, 0°):** the sparsest — **148** (94 raw), because here lay the contact base of the pear through which the pressure and heat flowed, leaving the smallest contraction during the rounding.
    - 6. **(2) The leading side (0°, 90°W): 174** (129 raw; 162 excluding the Orientale basin).
    - 7. **(3) The point opposite the Earth (0°, 180°): 221** (216 raw) — acted as a "free head" at the top of the dome and was not squeezed from outside.
    - 8. **(4–5) The north pole (90°N) and the south pole (90°S): 233** (227 raw) and **246** (211 raw) — the foci of maximum convergence during the rounding.
    - 9. **(6) The trailing side (0°, 90°E): 252** (253 raw) — combines the convergence of the ring with the gathering of ejecta that lagged behind in the rotation (Chapter 11).
10. **A blind fit without tuning parameters (Zero-Parameter Fit, §C.3):** when the average triaxial shape of the whole population (20+ km) is recovered from the ratios of the six axes (poles/sides = 1.12, poles/ends of the Earth axis = 1.30), an axis ratio of **1.37 : 1.15 : 1** results. A purely theoretical rounding of this shape into a sphere predicts a clean density ratio of **1.21** between the polar caps (30°) and the equatorial belt ($|\phi| < 15^\circ$) — an almost perfect match to the direct measurement on the surface (**1.19**, and in the full clean count **1.20 ± 0.05**) with no free parameter at all!

05 · THE KEY EQUATIONS

## Academic explanation and the key equations

The hydrostatic equilibrium equation of the Maclaurin spheroid (for eccentricity $e = \sqrt{1 - c^2/a^2}$ and rotation rate $\Omega = 2\pi/T$) and Roche's tidal stability criterion (k < 0.09) are expressed as follows (§C.1–C.2):

$$\frac{\Omega^2}{\pi G \rho} = \frac{2\sqrt{1-e^2}}{e^3}(3 - 2e^2)\arcsin(e) - \frac{6(1-e^2)}{e^2}$$

$$\rho_{\min} = \frac{\Omega^2}{0.4493\,\pi G}$$

$$k = \frac{\rho\, r^3}{M_E} < 0.09$$

$$R_{\text{Roche, rigid}} = 1.26\, R_{\oplus} \left( \frac{\rho_{\oplus}}{\rho} \right)^{1/3}$$

For a period of 5.5 hours, the lower stability limit on density is $\rho_{\min} = 1,070$ kg/m³ (at 1,000 kg/m³ there is no equilibrium solution and the body breaks apart), and at the detachment density (2,600 kg/m³) the Roche limit for a free fluid stands at 19,971–20,100 km — which shows that at the earlier detachment distance (16,400–16,610 km) the Moon held together thanks to **a rigid outer shell that had already frozen**!

06 · THE SIGNATURE

## The signature in reality: crater diameter as a "chronological clock" of the stages of rounding

Since craters kept landing on the Moon while it rounded from peak flattening (1.8 : 1.4 : 1) to the rigid sphere at detachment (≤ 1.04), the largest craters (formed near peak flattening) went through the full contraction of the poles, while the smaller, later craters went through only the end of the rounding (§A.8.5, §C.3):

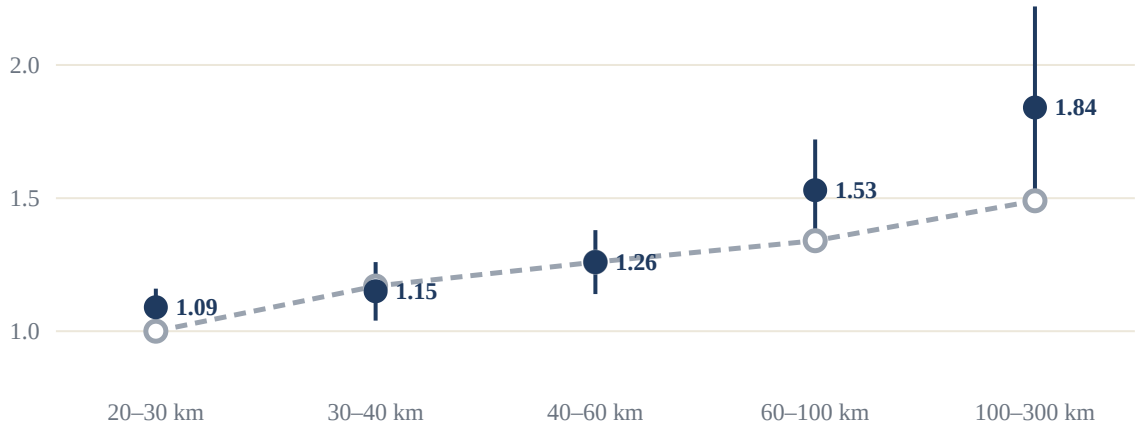
- **Craters of 100–300 km (peak flattening):** a measured pole/equator ratio of **1.84 ± 0.38** (prediction for peak flattening 1.8:1.4:1: **1.49**).
- **Craters of 60–100 km (crossing the corotation line):** a measured ratio of **1.53 ± 0.19** (prediction for the 1.40 shape: **1.34**).
- **Craters of 40–60 km (mid-rounding):** a measured ratio of **1.26 ± 0.12** (prediction for the 1.30 shape: **1.26** — an exact match!).
- **Craters of 30–40 km (end of the rounding):** a measured ratio of **1.15 ± 0.11** (prediction for the 1.20 shape: **1.17**).
- **Craters of 20–30 km (near and after detachment):** a measured ratio of **1.09 ± 0.07** (prediction for a spherical body 1.0: **1.00**).

## The lab box: the clean count by crater size, the six axes and the Maclaurin matrix (Paper 07, Parts A–C.2)

| Crater-diameter group / measure | Associated dynamical stage | Theoretical shape (a:b:c) | Predicted ratio (pole/equator) | Measured ratio in the clean count |
|---|---|---|---|---|
| **The whole population (≥ 20 km)** | **Combined average (six-axis recovery)** | **1.37 : 1.15 : 1** | **1.21 (no tuning!)** | **1.20 ± 0.05 (1.19 direct)** |
| **100–300 km** | Peak flattening (T = 5.5 h) | 1.8 : 1.4 : 1 | 1.49 | **1.84 ± 0.38** |
| **60–100 km** | **Crossing the corotation line** | **1.40** | **1.34** | **1.53 ± 0.19** |
| **40–60 km** | Middle of the inward rounding | 1.30 | 1.26 | **1.26 ± 0.12** |
| **30–40 km** | End of the rounding (2,600 kg/m³) | 1.20 | 1.17 | **1.15 ± 0.11** |
| **20–30 km** | Near and after detachment (spherical body) | 1.00 | 1.00 | **1.09 ± 0.07** |

| Axis end (30° cap) | Coordinates | Raw density (≥ 20) | Clean density (≥ 20) | Clean density (≥ 40) | Role in the triaxial structure |
|---|---|---|---|---|---|

| | | | | | |
|---|---|---|---|---|---|
| **1. Sub-Earth** | **0°, 0°** | **94** | **148** | **48** | **The global minimum — the contact base of the pear** |
| **2. Leading side** | 0°, 90°W | 129 | **174** (162 without Orientale) | 61 | A peripheral flank (hard head-on impacts) |
| **3. Anti-Earth** | 0°, 180° | 216 | **221** | 63 | A "free head" at the top of the dome (no partition) |
| **4. North pole** | **90°N** | **227** | **233 (300 in the 15° core)** | **89 (108 at 10°)** | **The focus of polar convergence (clean crust without SPA)** |
| **5. South pole** | **90°S** | **211** | **246 (305 in the 15° core)** | **80 (135 at 10°)** | **The focus of polar convergence (after removing the SPA core)** |
| **6. Trailing side** | 0°, 90°E | 253 | **252** | 92 | A ring peak + gathering of ejecta from behind |




Blue circle: measured in the clean count (with error) · grey circle: shape prediction

The pole/equator density ratio by crater diameter: the larger and older the crater, the more of the rounding it went through.

07 · NEXT

# The door ahead

Chapter 14

External Rotational Differentiation — Polar Contraction, the Thin Polar Crust and the Crater-Size Tests →

← Chapter 12 · Prologue

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